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Sonargaon University (SU)
সোনারগাঁও ইউনিভার্সিটি (এসইউ)

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LECTURE SHEET

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Chapter: Capital Budgeting Techniques

Questions at a glance:

1. What is Payback Period? State the advantages and disadvantages of payback period.
2. What is Accounting rate of return? Why ARR is a misleading measure?
3. What is Net Present Value? State the rationale for NPV.
4. What are the advantages and disadvantages of Net Present Value?
5. What is Internal Rate of Return? State the rationale for Internal Rate of return.
6. What are the advantages and disadvantages of internal Rate of Return?
7. What is profitability index? What are the advantages and disadvantages of profitability index?
8. What are the relation between NPV and PI? Why NPV is better than PI?
9. Describe the NPV Profile.
10. Which is the better assumption-that cash flows can be reinvested at the cost of capital, or that they can be reinvested at the project's IRR?
11. Why IRR is so popular?
12. Why NPV is better measure or technique among all?
13. Regular IRR vs. Modified IRR
14. What are the causes of arising conflict, if cash flow pattern are different?

1. *What is Payback Period? State the advantages and disadvantages of payback period.*

Answer: The Payback period of an investment project is the amount of time required for the firm to recover its initial investment in a project, as calculated from cash inflows.

Merits of payback period:

1. Popular technique in project evaluation: The payback period is very much popular technique in project evaluation due to its computational simplicity and intuitive appeal.

2. Considers cash flow: Payback period considers cash flows rather than accounting profit. Since profit can mislead the decision while cash flow does not do that, payback period helps the manager to take better decision.

3. Provides information about time required to recover initial investment: Payback period is the only way which gives the information about how much time is needed to recover the initial investment of a project.

4. Gives information about liquidity: It provides information about how long funds will be tied up in a project. Thus shorter the payback period others things remains constant, the greater the projects liquidity. So it gives information about liquidity.

5. Measures the risk of a project: Since cash flows expected in the distant future are generally riskier than near term cash flows, the payback period is often used as an indicator of a project's riskiness.

Demerits of payback period:

1. The major short coming of the payback period method is that it ignores cash flows that are paid or received after the payback period. Consequently it cannot be regarded as a measure of profitability. So it overlooks the objectives of firm's value maximization.

2. It fails to take fully into account the time factor in the value of money. Received sooner is better because of time value of money. It simply adds cash flows without regard to the timing of these f1ows.

3. The maximum acceptable payback period is merely a subjectively determine number. It does not give a definite investment signal.

2. *What is Accounting rate of return? Why ARR is a misleading measure? / Problems of ARR*

Answer: Accounting rate of return (ARR) examines a project's contribution to the firm's net income. It is also called average rate of return. Very few companies still use the ARR, and it really has no redeeming features.

$$\text{ARR} = \frac{\text{Average annual after tax profit}}{\text{Average investment}}$$

ARR is a misleading measure because –

1. It considers the profit instead of cash flows. As profit is a misleading term, so ARR can mislead.

2. AAR is not a true rate of return as it ignores time value of money.

3. It uses an arbitrary benchmark cutoff rate.

3. *What is Net Present Value? State the rationale for NPV.*

Answer: Net present value (NPV) is a sophisticated capital budgeting technique; found by subtracting a project's initial investment from the present value of its cash inflows discounted at a rate equal to the firm's cost of capital.

NPV = Present value of cash inflows - Initial investment.

Rationale for NPV: If the discount rate is chosen correctly then;

1. The rationale for NPV method is straightforward. An NPV of zero signifies that the project's cash flows arc exactly sufficient to repay the invested capital and to provide the required rate of return on that capital.

2. If a project has a positive NPV, then it is generating more cash than is needed to service the debt and to provide the required return to shareholders and this excess cash accrues solely to the firm's stockholders. Therefore, if a firm takes or a project with a positive NPV, the wealth of the stockholders increases.

3. NPV totally reflects on firm's value. How much is added to firm's value is known by NPV. No other techniques can give such information. Suppose; a project having NPV = $500 meaning that the project is able to meet its initial investment and would raise the firm's value by $500.

4. *What are the advantages and disadvantages of Net Present Value?*

Advantages of Net Present Value:

1. It considers the time value of money.

2. It is based on cash flows not profits.

3. It considers the whole life of the project; it does not ignore the cash flows as payback period method. For NPV, using the discounting factor, NPV covers the whole life.

4. It is an absolute measure of returns.

Disadvantages of Net Present Value:

1. It is very hard to explain to the managers since it requires an understanding of discounting.

2. It requires knowledge of cost of capital or cost of finance which is also not easy to estimate.

3. It is normally very hard to forecast or estimate future cash flows for investment evaluation.

5. *What is Internal Rate of Return? State the rationale for Internal Rate of return.*

Answer: The internal rate of return (IRR) is one of the most widely used capital budgeting techniques. Internal rate of return (IRR) is the discount rate that equates the present value of a project's expected cash inflows to the present value of the project's costs. It is the rate of return that the firm will earn if it invests in the project and receives the given cash inflows.

Rationale for internal rate of return:

Why is the particular discount rate that equates a project's cost with the present value of its receipts (the IRR) so special? The reason is based on this logic:

1. The IRR on a project is its expected rate of return. .

2. If the internal rate of return exceeds the cost of the funds used to finance the project, a surplus will remain after paying for the capital, and this surplus will accrue to the firm's stockholders.

3. If IRR is greater than its cost of capital, then taking on a project will increase shareholders' wealth. On the other hand, if the IRR is less than the cost of capital, then taking on the project will impose a cost on current stockholders. It is this "breakeven" characteristic that makes the IRR useful in evaluating capital projects.

6. *What are the advantages and disadvantages of internal Rate of Return?*

Advantages of Internal Rate of return:

1. It considers time value of money by discounting cash flows to the present values.

2. It uses cash flows not profits; profits may be determined subjectively.

3. It considers the whole life of the project.

4. It is expressed in percentage form and therefore easily understood.

Disadvantages of Internal rate of Return:

1. Interpolation and use of the formula provides an estimate and therefore IRR the accuracy can be doubted

2. It is fairly complicated to calculate.

3. It is not a measure of absolute figures but relative figures which may not be easily understood if details are not given.

4. It produces multiple rates which may be confusing for taking decisions.

5. It assumes that intermediate cash flows arc reinvested at the internal rate of return.

7. *What is profitability index? What are the advantages and disadvantages of profitability index?*

Answer: Profitability index is defined as the present value of the future cash flows divided by the initial investment.

Advantages of profitability index:

1. PI is closely related to NPV, generally leading to identical decisions.

2. It is easy to understand and communicate.

3. It may be useful when available investment funds are limited.

Disadvantages of profitability index:

1. PI may lead to incorrect decisions in comparisons of mutually exclusive investments.

2. It cannot give the information about how much is added to firm's value.

8. *What are the relation between NPV and PI? Why NPV is better than PI?*

Answer: If a project has a positive NPV, then the present value of the future cash flows must be bigger than the initial investment. The profitability index would thus be bigger than 1 for a positive NPV investment and less than 1 for a negative NPV investment. For example, If a project costs $200 and the present value of its future cash flows is $220, the profitability index value would be $220/200 = 1.1. Notice that the NPV for this investment is $20, so it is a desirable investment.

NPV is better than PI because-

1. PI is a relative measure cannot say how much rate would be added to the firm, expresses is only the relative profitability.

2. NPV is an absolute measure which clearly shows how much value would be added to the firm.

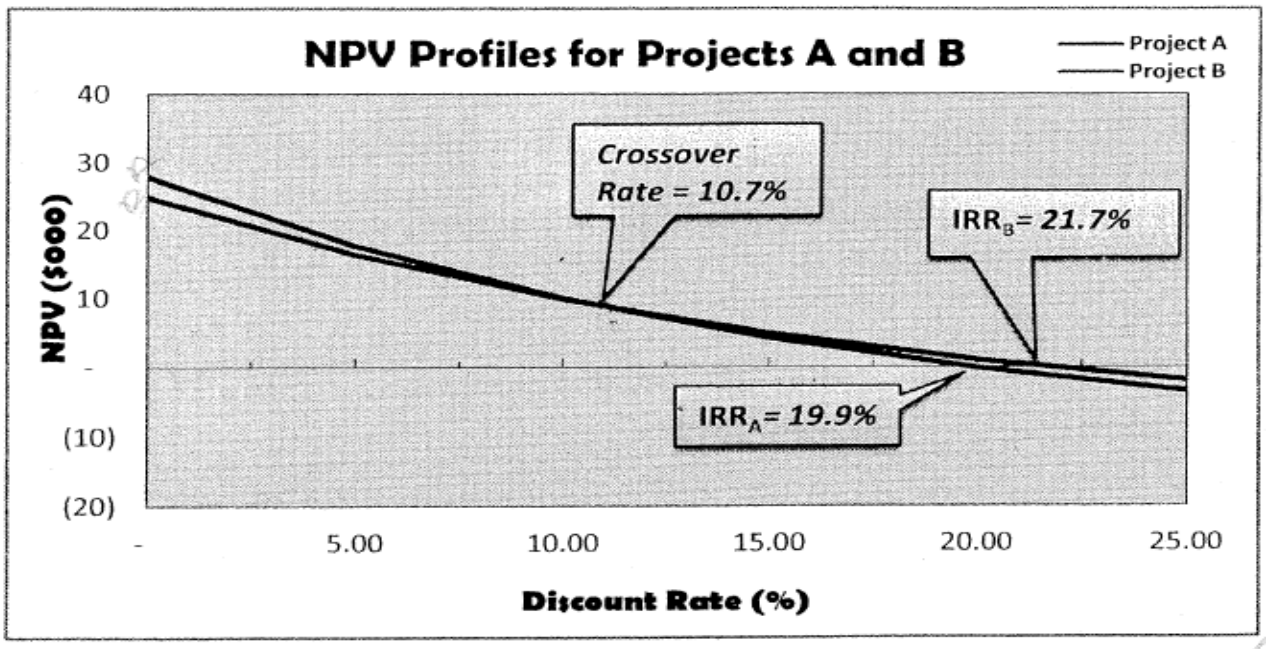
9. *Describe the NPV Profile.*

NPV profiles: NPV profile is a curve showing the relationship between a projects' NPV and the firms' cost of capital or discount rate. These profiles are useful in evaluating and comparing projects, especially when conflicting ranking exist. They are best demonstrated via an example.

The three sets of coordinates for projects A and B and their NPV values are given below –

Discount Rate (%)	NPV	
	Project A	Project B
0	28,000	25,000
10	11,071	10,924
19.9	0	-
21.7	-	0

Plotting the data, from the above Table, results in the NPV profiles for projects A and B shown in below Figure:



The figure reveals three important facts:

1. The IRR of project B is greater than the IRR of project A, so managers using the IRR method to rank projects will always choose B over A if Both projects are acceptable.

2. The NPV of project A is sometimes higher and sometimes lower than the NPV of project B; thus, the NPV method will not consistently rank A above B or vice versa. The NPV ranking will depend on the firm's cost of capital.

3. When the cost of capital is approximately 10.7%, project A and B have identical NPVs.

Crossover rate is the cost of capital at which the net present values of two projects are equal. It is the point at which the net present value profile of one project crosses **over** (intersects) the net present value profile of the other project.

Three basic conditions can cause NPV profiles to cross, and thus conflicts to arise between NPV and IRR methods –

1. When project size (or scale) differences exist, meaning that the cost of one project is larger than that of the other, or;

2. When timing differences exist, meaning that the timing of cash flows from the two projects differs such that most of the cash flows from one project come in the early years while most of the cash flows from the other project come in the later years.

3. When project life or unequal lives exist, meaning that two projects are mutually exclusive, but having different project's life- one is 3 years while another is 5 years.

10. *Which is the better assumption-that cash flows can be reinvested at the cost of capital, or that they can be reinvested at the project's IRR?*

The best assumption is that projects' cash flows can be reinvested at the cost of capital, which means that the NPV method is more reliable.

We should reiterate that, when projects are independent, the NPV and IRR methods both lead to exactly the same accept-reject decision. However, when evaluating mutually exclusive projects, especially those that differ in scale and/or timing, the NPV method should b used.

11. *Why IRR is so popular?*

Answer: The internal rate of return (IRR) is a capital budgeting metric used by firms to decide whether they should make investments.

1. It is an indicator of the efficiency of an investment, as opposed to net present value (NPV), which indicates value or magnitude.

2. The IRR is the annualized effective compounded return rate which can be earned on the invested capital, i.e., the yield on the investment.

3. A project is a good investment proposition if its IRR is greater than the rate of return that could be earned by alternative investments. Thus, the IRR should be compared to an alternative cost of capital including an appropriate risk premium.

4. Mathematically the IRR is defined as any discount rate that results in a net present value of zero of a series of cash flows.

5. In general, if the IRR is greater than the project's cost of capital, or hurdle rate, the project will add value for the company.

12. *Why NPV is better measure or technique among all?*

Answer: On a purely theoretical basis, NPV is the better approach because:

1. NPV assumes that intermediate cash flows are reinvested at the cost of capital whereas IRR assumes they are reinvested at the IRR.

2. Certain mathematical properties may cause a project with non-conventional cash flows to have zero or

more than one real IRR; this problem does not occur with the NPV approach.

3. Despite its theoretical superiority, however, financial managers prefer to use the IRR because of the preference for rates of return.

13. *Regular IRR vs. Modified IRR*

Answer: The differences between regular IRR and modified IRR are as follows:

Points	Regular IRR	Modified IRR
1. Definition	It is the interest rate that makes the NPV equal to zero for the series of cash flows.	It is the interest rate that equates the project's cost to the terminal value at the end of the project.
2. Assumptions	The regular IRR assumes that the cash flows from each project's are re-invested at the project's own IRR.	The modified IRR assumes that positive cash flows are reinvested at the firm's cost of capital, and the initial outlays are financed at the firm's financing cost.
3. Accuracy	It does not accurately reflect the cost and profitability of a project.	MIRR more accurately reflects the cost and profitability of a project.

14. *What are the causes of arising conflict, if cash flow pattern are different?*

Answer: There are following two reasons –

1. Reinvestment Assumption: One underlying cause of conflicting rankings is different implicit assumptions about the reinvestment of intermediate cash inflows – cash inflows received prior to the termination of a project. NPV assumes that intermediate cash inflows are reinvested at the cost of capital, whereas IRR assumes that intermediate cash inflows are reinvested at a rate equal to the project's IRR. To eliminate the reinvestment rate assumption of the IRR, some practitioners calculate the modified internal rate of return (MIRR). The MIRR is found by converting each operating cash inflow to its future value measured at the end of the project's life and then summing the future values of all inflows to get the project's terminal value. Each future value is found by using the cost of capital, thereby eliminating the reinvestment rate criticism of the traditional IRR. The MIRR represents the discount rate that causes the terminal value just to equal the initial investment. Because it uses the cost of capital as the reinvestment rate the MIRR is generally viewed as a better measure of a project's true profitability than the IRR.

2. Magnitude and timing of cash flows: The greater the difference between the magnitude and timing of cash inflows, the greater the likelihood of conflicting rankings. Conflicts based on NPV and IRR can be reconciled computationally; to do so, one creates and analyzes an incremental project reflecting the difference in cash flows between the two mutually exclusive projects.

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Math Problems and Solution

1. You are a financial analyst for Damon Electronics Company. The director of capital budgeting has asked you to analyze two proposed capital investments, Projects X and Y. Each project has a cost of $10,000 and the cost of capital for each project is 12 percent. The projects' expected net cash flows are as follows:

Year	Project X	Project Y
0	($10,000)	($10,000)
1	6,500	3,500
2	3,000	3,500
3	3,000	3,500
4	1,000	3,500

a. Calculate each project's payback period, discounted payback period, net present value (NPV), internal rate of return (IRR) and modified internal rate of return (MIRR).

b. Which project or projects should be accepted if they are independent?

c. Which project should be accepted if they are mutually exclusive?

Solution:

a.

Payback Period (PBP)

Project X:

Year	Cash Flow	Cumulative Cash Flow
1	6,500	6,500
2	3,000	9,500
3	3,000	12,500
4	1,000	13,500

$$PBP = \text{Year before full recover} + \frac{\text{Unrecover Cost}}{\text{Cash inflow during the year}}$$

$$= 2 + \frac{10,000 - 9,500}{3,000}$$

$$= 2.17 \text{ years}$$

Project Y:

$$PBP = \frac{\text{Initial investment}}{\text{Annual equal cash inflow}}$$

$$= \frac{10,000}{3,500}$$

$$= 2.86 \text{ years}$$

Discounted Payback Period

Project X

$\frac{CF}{(1+.12)^n}$

Year	Cash Flow	Discounted cash flow @ 12%	Cumulative Discounted Cash Flow
1	6,500	5,803.57	5,803.57
2	3,000	2,391.58	8,195.15
3	3,000	2,135.34	10,330.49
4	1,000	635.52	10,966.01

$$PBP = 2 + \frac{10,000 - 8,195.15}{2,135.34}$$

$$= 2.85 \text{ years}$$

Project Y

$\frac{CF}{(1+.12)^n}$

Year	Cash Flow	Discounted Cash Flow @ 12%	Cumulative Discounted Cash Flow
1	3,500	3,125	3,125
2	3,500	2,790.18	5,915.18
3	3,500	2,491.23	8,406.41
4	3,500	2,224.31	10,630.72

$$PBP = 3 + \frac{10,000 - 8,406.41}{2,224.31}$$

$$= 3.72 \text{ years}$$

Net Present Value (NPV)

Project x

NPV = PV of cash inflow − Initial investment

$$= \frac{6,500}{(1+.12)^1} + \frac{3,000}{(1+.12)^2} + \frac{3,000}{(1+.12)^3} + \frac{1,000}{(1+.12)^4} - 10,000$$

$$= 5,803.57 + 2,391.58 + 2,135.34 + 635.52 - 10,000$$

$$= 10,966.01 - 10,000$$

$$= 966.01$$

Project Y

$$NPV = A\left[\frac{1-(1+i)^{-n}}{i}\right] - \text{Initial investment}$$

$$= 3,500\left[\frac{1-(1+.12)^{-4}}{.12}\right] - 10,000$$

$$= 10,630.72 - 10,000$$

$$= 630.72$$

Internal Rate of Return (IRR)

Project x

At 20%,

$$NPV = \frac{6,500}{(1+.20)^1} + \frac{3,000}{(1+.20)^2} + \frac{3,000}{(1+.20)^3} + \frac{1,000}{(1+.20)^4} - 10,000$$

$$= 5,416.67 + 2,083.33 + 1,736.11 + 482.25 - 10,000$$

$$= 9,718.36 - 10,000$$

$$= -281.64$$

$$IRR = LR + \frac{NPV_{LR}}{NPV_{LR} - NPV_{HR}} (HR - LR)$$

$$= 12\% + \frac{966.01}{966.01 - (-281.64)} (20\% - 12\%)$$

$$= 12\% + 6.19\%$$

$$= 18.19\%$$

Project y

At 16%,

$$NPV = 3,500 \left[\frac{1 - (1+.16)^{-4}}{.16} \right] - 10,000$$

$$= 9,793.63 - 10,000$$

$$= -206.37$$

$$IRR = LR + \frac{NPV_{LR}}{NPV_{LR} - NPV_{HR}} (HR - LR)$$

$$= 12\% + \frac{630.72}{630.72 - (-206.37)} (16\% - 12\%)$$

$$= 12\% + 3.01\%$$

$$= 15.01\%$$

Modified Internal Rate of Return (MIRR)

$CF(1+.12)^{4-n}$

	Cash Flow		Future Value @ 12%	
	Project x	Project y	Project x	Project y
1	6,500	3,500	9,132.03	4,917.25
2	3,000	3,500	3,763.20	4,390.40
3	3,000	3,500	3,360	3,920
4	1,000	3,500	1,000	3,500
FV of cash inflow end of year 4			17,255.23	16,727.65

Project x

$$PV \text{ of costs} = \frac{\text{Terminal Value}}{(1+MIRR)^n}$$

$$\Rightarrow 10,000 = \frac{17,255.23}{(1+MIRR)^4}$$

$$\Rightarrow (1+MIRR)^4 = \frac{17,255.23}{10,000}$$

$$\Rightarrow (1+MIRR)^4 = 1.725523$$

$$\Rightarrow (1+MIRR) = \sqrt[4]{1.725523}$$

$$\Rightarrow 1+MIRR = 1.1461$$

$$\Rightarrow MIRR = 1.1461 - 1$$

$$\Rightarrow MIRR = .1461$$

$$\therefore MIRR = 14.61\%$$

Project y

$$PV \text{ of costs} = \frac{\text{Terminal Value}}{(1+MIRR)^n}$$

$$\Rightarrow 10,000 = \frac{16,727.65}{(1+MIRR)^4}$$

$$\Rightarrow (1+MIRR)^4 = \frac{16,727.65}{10,000}$$

$$\Rightarrow (1+MIRR)^4 = 1.672765$$

$$\Rightarrow 1+MIRR = \sqrt[4]{1.672765}$$

$$\Rightarrow 1+MIRR = 1.1373$$

$$\Rightarrow MIRR = 1.1373 - 1$$

$$\Rightarrow MIRR = .1373$$

$$\therefore MIRR = 13.73\%$$

b.

Both projects should be accepted if they are independent. Because both projects are acceptable under the PBP, NPV, IRR and MIRR criteria.

c.

All methods rank Project X over Project Y. We would choose project X if they are mutually exclusive.